■■ AIDAG-WIND — System Architecture

Layer-1 DAG AI Infrastructure · Built with Go & Python · Testnet Live · Proof-First

[1] Network & Event Ingestion Layer

where signals land & passive listening occurs

- **UDP Sentinel Port (9001):** Sub-millisecond UDP socket capture for wind signals.
- **Zero-CPU Idle Sentinel:** Go-based lightweight daemon operating near 0% CPU footprint.
- **Network Isolation:** Strict internal Docker bridge (cell_network) restricting public vector access.

[2] Autonomous AI Execution Engine

on-demand neural inference & threat analysis

- **Vector Analysis Core:** Python 3.11 inference engine processing micro-vectors.
- **Threat Classifier:** Active anomaly detection distinguishing valid payloads from malicious vectors.
- **Proof Generator:** SHA-256 state proof generator outputting cryptographic execution tokens.

[3] Verifiable Consensus & Minting Layer

the immutable proof loop

- **HTTP Proof Bridge (9002):** Internal state synchronization handshake.
- **AIDAG-MINT Engine:** Immediate ledger stamping of verified execution tokens.
- **Execution Verification:** Unalterable block/vertex status registration (0x-AIDAG-PROOF).

[4] Security & Containment Shield

zero-trust sandboxed isolation

- **Zero-Attack-Surface:** Only Go Sentinel exposed publicly; AI agent completely sandboxed.
- **Containment Protocol:** AI engine failures do not disrupt core DAG consensus operations.
- **Anomaly Rejection:** Deterministic rejection of malicious payload vectors before block minting.

■ Live Production Execution Proofs & Benchmark Validation

```
[TEST 1: VALID VECTOR]
■ [SIGNAL] Wind Vector Caught: AIDAG_TX_VALID_5500
■ [AIDAG-MINT] Proof Stamped: 0x-AIDAG-D087B577D0F6 | Status: VERIFIED_VALID

[TEST 2: MALICIOUS ATTACK MITIGATION]
■ [SIGNAL] Wind Vector Caught: AIDAG_TX_ATTACK_MALICIOUS_99999999
■ [AIDAG-MINT] Proof Stamped: 0x-AIDAG-D7D411AAA953 | Status: REJECTED_ANOMALY
```